

[RI/2024/Q1]

1 Name your Jupyter Notebook as:

```
TASK1_<your name>_<class>_<index number>.ipynb
```

An online shopping platform stores its data in a MongoDB database.

The platform needs to store information about products, customers, and orders.

Your task is to perform several operations on the database.

For each of the sub-tasks, add a comment statement at the beginning of the code using the hash symbol '#' to indicate the sub-task the program code belongs to, for example:

```
In [1]: #Task 1.1
        Program code
```

Output:

Task 1.1

Write Python code to create a MongoDB database named `online_shop`. Ensure that this database is empty before proceeding.

Three JSON files are provided, each containing multiple records to be inserted into specific collections within the database. The files are `PRODUCTS.json`, `CUSTOMERS.json` and `ORDERS.json`.

Create collections named `products`, `customers`, and `orders`, and insert the data from each JSON file into the corresponding collection: [3]

JSON file	Insert into collection	Sample JSON record
PRODUCTS	products	{ "product_id": 1, "name": "Laptop", "model": "XPS 13", "category": "Electronics", "price": 1200, "stock": 30 }
CUSTOMERS	customers	{ "customer_id": 1, "name": "Alex", "email": "alexB@abc.com", "phone": "12345678" }
ORDERS	orders	{ "order_id": 1, "customer_id": 1, "products": [{"product_id": 2, "quantity": 1}], "order_date": "2024-01-15" }

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Task 1.2

Add to your program code to retrieve and display all products in the `Books` category from the `products` collection.

Use the `pprint` module to display your data in a readable format:

```
from pprint import pprint
pprint(your_data)
```

Run your program. [2]

Task 1.3

Add to your program code to find all orders placed by the customer named `Charles`. Assume there is only one customer with this name.

Display the details of these orders from the `orders` collection.

Use `pprint` to display your data in a readable format.

Run your program. [4]

Task 1.4

Add to your program code to calculate the revenue generated from all orders.

Display the revenue in a tabular format with headers for Order ID and Revenue.

For each order, compute the revenue as the sum of the price of each product multiplied by the quantity ordered.

Run your program. [5]

Save your Jupyter Notebook for Task 1.